

Spatio-temporal validation of two mechanistic models of *Aedes aegypti* population dynamics in ten Argentine localities

Tomás V. San Miguel^{1,2}, Daniele Da Re³, Manuel O. Espinosa⁴,
María V. Periago^{1,4}, Laura Lopez⁵, Claudio D. Guzmán⁵, Verónica
Andreo^{1,2}

[1] Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET), Buenos Aires, Argentina

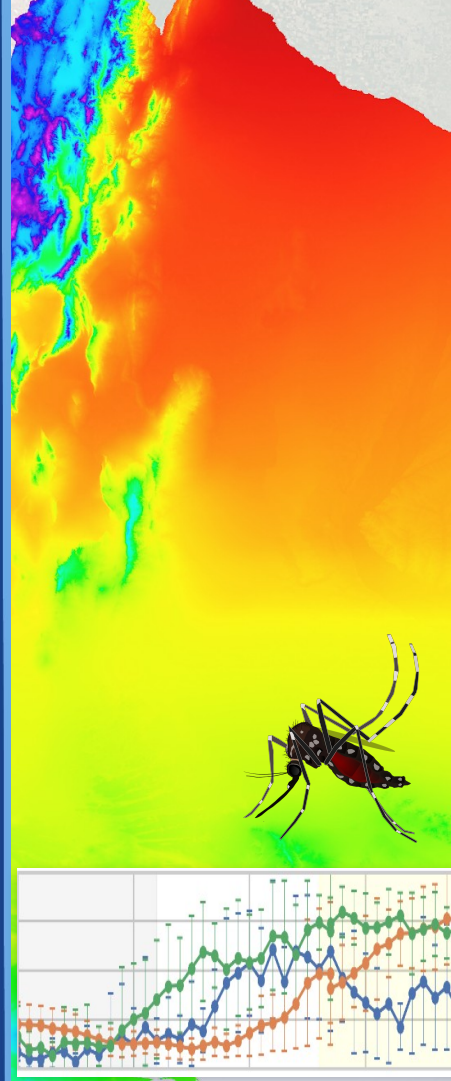
[2] Instituto de Altos Estudios Espaciales "Mario Gulich" (UNC-CONAE), Falda del Cañete, Córdoba, Argentina

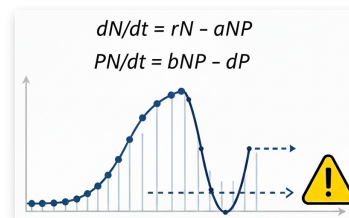
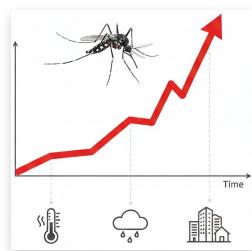
[3] Research and Innovation Centre, Fondazione Edmund Mach, San Michele all'Adige, Trento, Italia

[4] Fundación Mundo Sano, Buenos Aires, Argentina

[5] Programa de Zoonosis, Área de Epidemiología, Ministerio de Salud, Córdoba, Argentina

Gulich





Mechanistic models

Description of a system through the underlying mechanisms that govern it



Why validating?

- **Robustness:** do they work the same in different places or times?
- **Differences and similarities:** does it make a difference to use one model or another?
- **Stability/sensitivity:** how do they behave with changing conditions?
- Do they reflect reality?

Compare the predictive performance of two mechanistic models of *Aedes aegypti* population dynamics against ovitrap data in 10 Argentine locations for the period 2015-2024.

Características	Aguirre et al. (2021)	dynamAedes
Model type	Deterministic	Stochastic
Compartments	5: egg, larvae, pupae, adults 1 & adults 2	3: egg, juvenile, adults
Scale	Spatial: point; temporal: daily	
Inputs	Biological parameters, temperature, precipitation, relative humidity (historical and forecasted)	Biological parameters, temperature, initial liters of water
Programming language	C++ & Python	R (package dynamAedes)
Motivation	Develop a proactive system to monitor populations. Demo: https://aplicaciones.culich.unc.edu.ar/app/	Evaluate the potential for the species to re-invade Europe

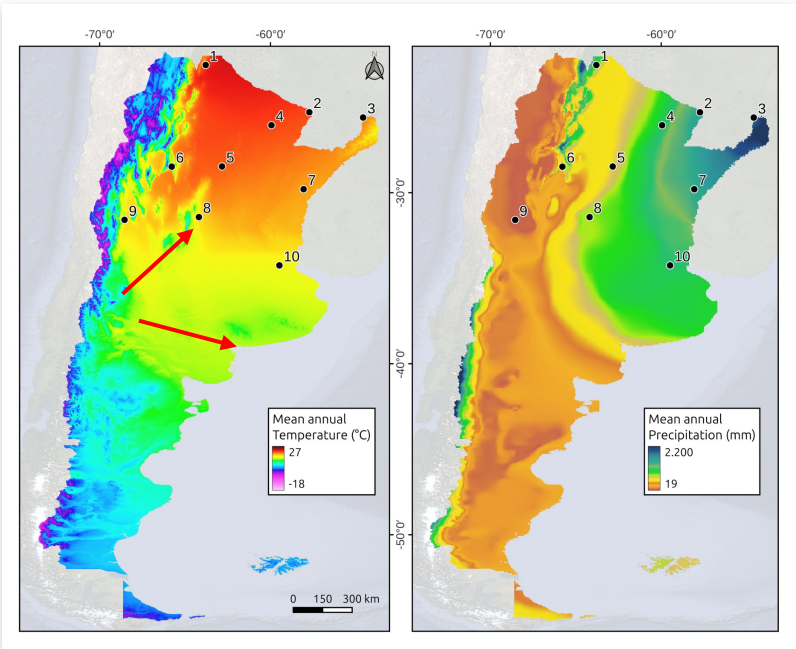


Aguirre et al. (2021)



Da Re et al. (2021), Da Re et al. (2022)

Ten locations selected because of the availability of ovitrap data collected by the Ministry of Health of Córdoba (8) and by Fundación Mundo Sano (1-7,9,10).



Locality, province (Lat, Long)	Climate type	Ovitrap	
		Number*	Period
1) Tartagal, Salta (22.5°S, 63.7°W)	Subtropical mountainous (Warm)	128	1-2016 to 52-2023
2) Clorinda, Formosa (25.2°S, 57.7°W)	Subtropical without dry season (Warm)	60	
3) Puerto Iguazú, Misiones (25.5°S, 54.5°W)		140	
4) Pampa del Indio, Chaco (26.0°S, 59.9°W)	Subtropical with dry season (Warm)	40	14-2019 to 52-2023
5) Añatuya, Santiago del Estero (28.4°S, 62.8°W)		50	
6) San Fernando del V. de C., Catamarca (28.4°S, 65.7°W)	Mountains (Arid)	206	51-2019 to 17-2023
7) Curuzú Cuatiá, Corrientes (29.7°S, 58.0°W)	Subtropical without dry season (Warm)	40	41-2018 to 35-2020
8) Córdoba, Córdoba (31.4°S, 64.1°W)	Pampean (Temperate)	300	40-2017 to 52-2019
9) Rawson, San Juan (31.5°S, 68.5°W)	Mountains (Arid)	40	44-2020 to 16-2023
10) San Antonio de Arco, Buenos Aires (34.2°S, 59.4°W)	Pampean (Temperate)	*2 per household 54	41-2020 to 52-2023

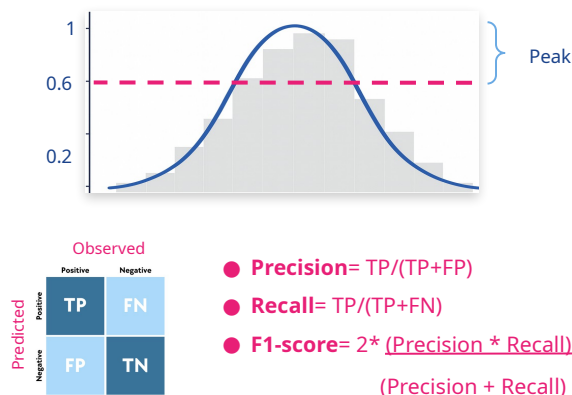
- **Simulation period:** 1 November 2015 to 31 December 2024.
- **Initial population:**
 - 1,000 eggs.
 - Aguirre: 1,000 1-litre containers. DynamAedes: 1,000 litres of water.
- **Daily meteorological data:**
 - NCEP GDAS/FNL 0.25° from NOAA (average temperature and relative humidity);
GPM_3IMERGDL 0.1° from NASA (precipitation).
- Kept **number of eggs** for the comparisons
- Number of eggs were **averaged weekly and normalized** between 0-1 for annual winter-winter cycles.

Models validation

Overall spatio-temporal performance

- RMSE and Spearman Correlation coefficient.
- Fit LMMs to estimate the weekly error marginal means (EMM) for each model and model-locality

Peak detection



Binomial GLMMs:

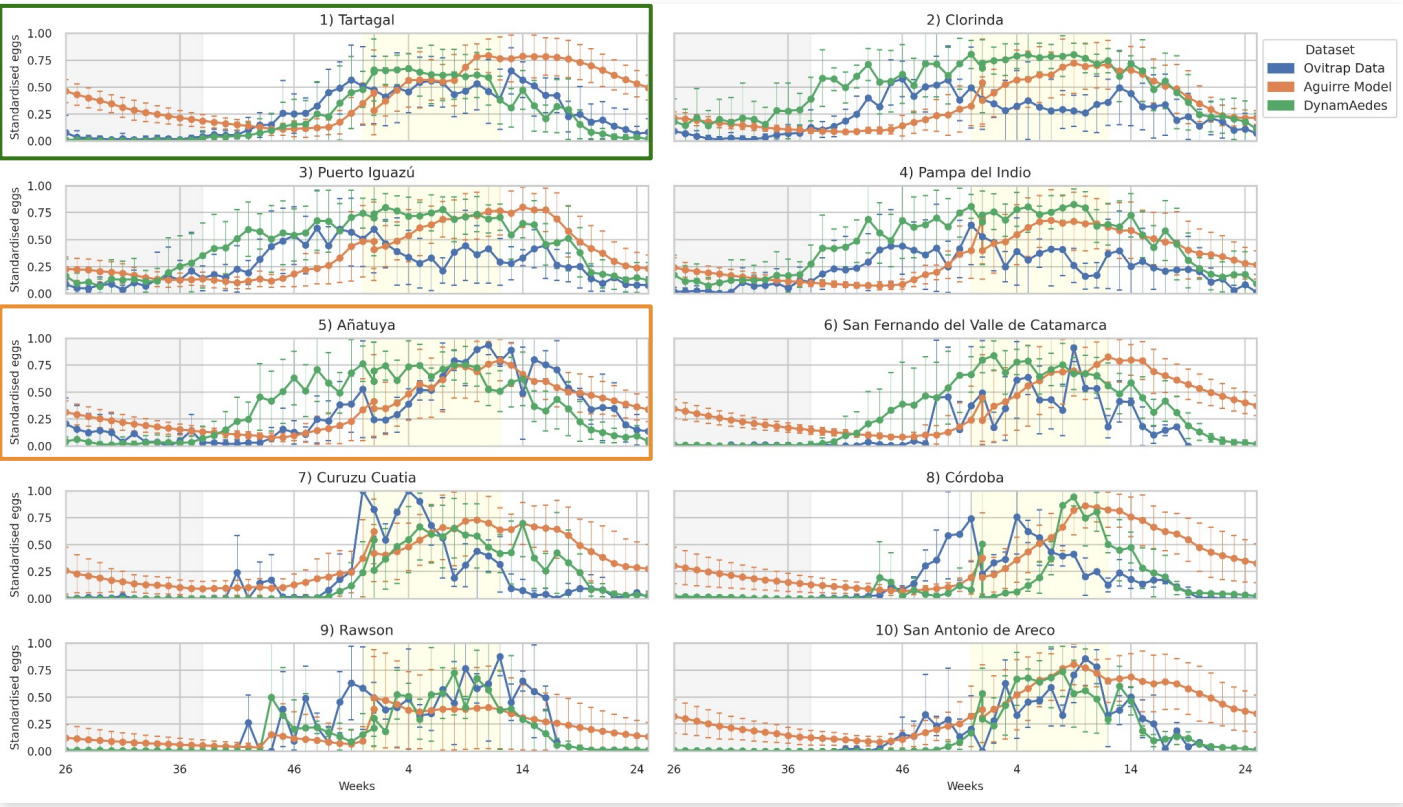
- Recall (1 = TP; 0 = FN) and Precision (1 = TP; 0 = FP)
- Fixed effect: models
- Random effects: locality, annual cycle

Seasonal pattern/phenology



- Season: ≥ 6 consecutive weeks with values > 0.2 , allowing interruptions of up to 2 weeks.
- Friedman test for median comparisons; Wilcoxon-Holm post-hoc tests for pairwise comparisons.

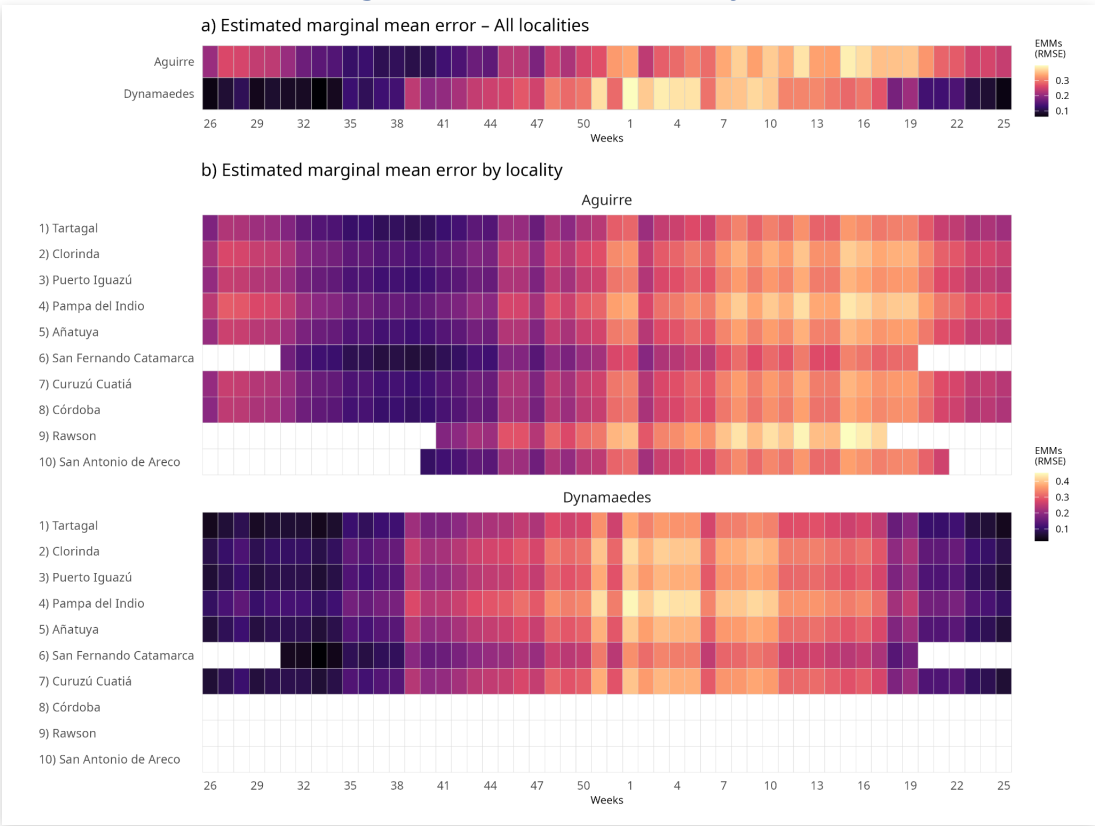
Egg abundance normalized by week for each locality.



- **Global RMSE**
dynamAedes: 0.11 - 0.29.
Aguirre: 0.12 - 0.34.
- **Spearman correlation**
dynamAedes: 0.62 - 0.92*.
Aguirre: 0.38 - 0.87**.
- **Best performances**
dynamAedes: Tartagal (RMSE: 0.11 y Corr: 0.92)
Aguirre: Añatuya (RMSE: 0.12 y Corr: 0.86).

* 9/10 localities p<0.05
** 8/10 localities p<0.05

Estimated Marginal Means of the weekly error

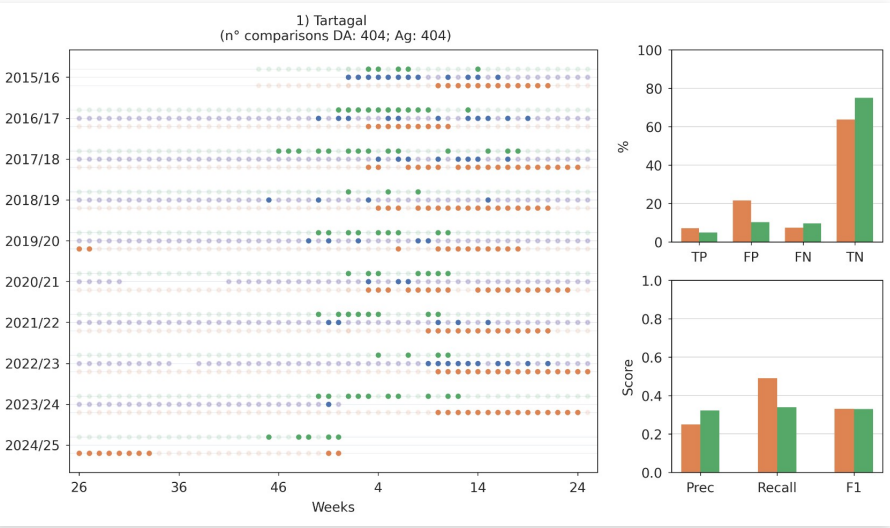


Larger errors appear in weeks with higher temperatures (greater egg abundance) but shifted among models.

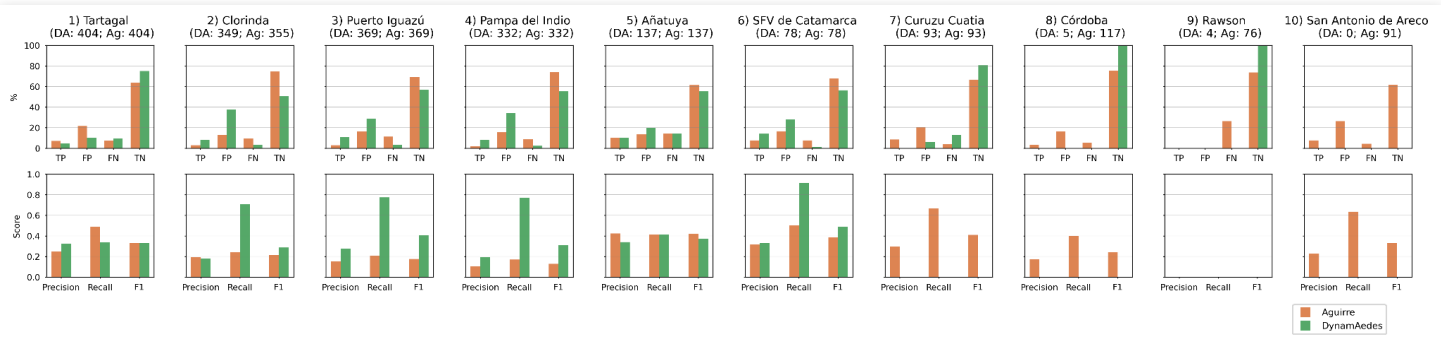
Errors were smaller in dynamAedes but increased earlier in the season, while Aguirre delayed the error increase but it remained more sustained over the weeks (smooth decline).

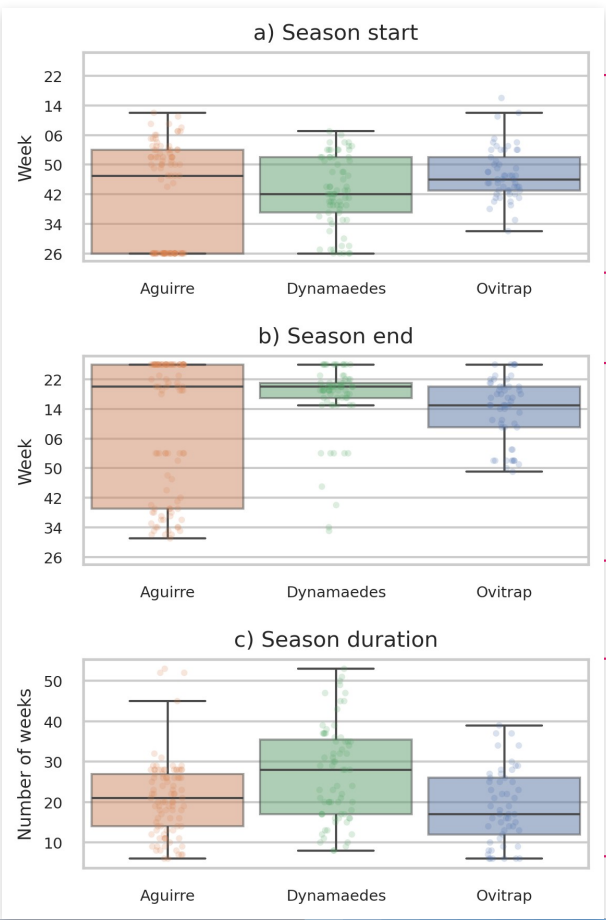
There were more early extinctions with dynamAedes than with Aguirre's:

- dynamAedes: 3 southernmost localities.
- Aguirre: Rawson.



- Both models overestimated the #weeks classified as peaks: 29.7% dynamAedes, 20.9% Aguirre, and 14.3% ovitraps.
- Both models correctly identified non-peak weeks (TN values were overall high).
- dynamAedes detected a significantly higher number of actual peaks compared to Aguirre (Recall of 43.5% vs. 27.2%).
- Both models were equally bad in predicting false peaks (Precision 6.7% Aguirre vs. 8.2% dynamAedes).





- dynamAedes anticipated the onset (W 42), while Aguirre was apparently closer to the ovitraps (W 47 vs 46) but highly variable.
- Both models differed significantly from the ovitraps, but not from each other.
- Aguirre and dynamAedes predicted the end of the season by W 20, while the median for the ovitraps was W 15.
- No differences were found among the three datasets.
- DynamAedes extended the duration observed in the ovitraps (17) by 11 weeks, and Aguirre by 4 weeks.
- No significant differences between Aguirre and ovitraps.

- **Variable overall performance**

- The RMSE range was similar for the 2 models → the average RMSE across sites is slightly better with dynamAedes.
- dynamAedes showed higher correlation with ovitrap data.
- No clear performance pattern according to climate or latitude.

- **dynamAedes predicted extinctions where established *Aedes aegypti* populations exist** (performs bad in “colder” places?).

- **dynamAedes was more accurate in identifying peaks** when they actually occurred.

- Both models predicted the end of the seasons well but **failed at the onset**. Aguirre's model was closer in terms of median onset week, but it was highly variable. **The onset is indeed variable!**

- **Aguirre's model better matched the observed season durations** (mostly because it tends to decrease very slowly?) → models' functional form is relevant

Comparison and Validation

It is necessary to compare models' performance in space and time to understand the uncertainty of their results and their limitations.

Purpose and objectives

Each model is developed with a specific purpose, which may or may not align with others.

Need for data

Field data are required for validation.
Laboratory data are required to quantify the mechanisms & update parameters.

Nature of models

They are abstractions of complex systems and will not be perfect.

Thank you



Dr. Verónica Andreo
Instituto Gulich
veronica.andreo@ig.edu.ar

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Threshold selection using the first stable plateau method

